Price Path Embeddings in a Hang Seng Index Portfolio

Samson Lam

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Abstract

This study explores whether embeddings of the recent trading history of a stock carry predictive information, and what they deliver in a portfolio. For each Hang Seng Index (HSI) stock, the daily return path and technical indicators are embedded in low dimensions, linearly by principal component analysis (PCA) and nonlinearly by a variational autoencoder (VAE). A pooled regression maps the embedding to a forecast of next-month returns. In a long-only portfolio, the forecast selects the stocks, and a Black-Litterman model constructs the weights. Listed HSI derivatives enhance yield through delta-hedged short calls and a put spread collar held at high volatility. From September 2021 to August 2026, the forecast has a rank information coefficient near 0.07, and the VAE and PCA perform similarly. The PCA portfolio with both derivatives strategies returns 6.28% a year against 3.09% for the HSI total return, an active return of 3.05% with an information ratio of 0.46. Stock selection adds 2.93% a year, and derivatives add 0.98%. Although the forecast itself is significant, its transfer into active return is not, and neither are the derivatives strategies.[1]

1 Introduction

Returns over the past year are known to predict returns over the following months (Jegadeesh and Titman, 1993), yet a year of daily returns and technical indicators holds far more inputs than a regression on noisy monthly returns can use. An embedding represents such a high-dimensional input in a few dimensions. It can be linear, by PCA, or nonlinear, by a VAE with long short-term memory (LSTM) layers. This study asks (i) whether embeddings of the past year of returns and technical indicators of HSI stocks carry information about their returns over the next month, (ii) whether a nonlinear embedding improves on a linear one, and (iii) what the embeddings deliver in a portfolio.

This study examines a USD 100 million long-only portfolio of HSI stocks with listed HSI derivatives, benchmarked against the HSI total return (HSI TR). The outperformance is decomposed into three sources: stock selection using an embedding forecast, portfolio construction that turns selection into weights, and yield enhancement by derivatives strategies.

Machine learning has been widely used to forecast stock returns from past prices. Gu et al. (2020) compare linear and nonlinear methods for predicting the returns of US stocks, trace the gains of trees and neural networks to nonlinear interactions among predictors, and find momentum among the dominant signals. Jiang et al. (2023) train a convolutional network on images of past price charts and find price patterns that differ from momentum and reversal and predict returns more accurately. Fischer and Krauss (2018) train an LSTM network (Hochreiter and Schmidhuber, 1997) on the return sequences of all S&P 500 constituents to predict whether a stock will beat the cross section on the next day, and Wu et al. (2026) feed daily features built from prices and volume to an LSTM in trend-following strategies. A VAE (Kingma and Welling, 2014) embeds such an input in a latent space of low dimension whose coordinates can serve as forecasting variables, and PCA provides its linear counterpart.

For portfolio construction, the mean-variance model of Markowitz (1952) suffers from extreme and unstable weights. Black and Litterman (1992) proposed instead to start from the expected returns implied by the market portfolio in equilibrium and to tilt them toward the investor's

[1] Code and replication data are available at *GitHub*.

views in proportion to the confidence attached to each view, and He and Litterman (1999) set the uncertainty of each view proportional to the variance of the prior, which removes the need to choose the scale of the prior. This study uses the model to turn the forecasts into weights.

The evidence on option strategies for equity portfolios is mixed. Israelov and Nielsen (2015) decompose a covered call into an equity position, a short volatility position and an equity timing position, and find that the first two are compensated while the timing position is not. The short volatility position is compensated by the volatility risk premium, which Bakshi and Kapadia (2003) infer from the negative average gain of a delta-hedged long option. On protection, Israelov (2019) finds protective puts less effective at reducing drawdowns than a static reduction of exposure, and Israelov and Klein (2016) find that collars earn less equity premium than their index and have underperformed it, while Szado and Schneeweis (2010) report that a collar adjusted by momentum, volatility and economic signals outperforms a passive one. Following this evidence, the yield strategy of this study hedges the delta of its short calls, and the protection strategy holds a collar only when implied volatility is high.

Applied to HSI stocks from September 2021 to August 2026, the study answers the three questions as follows. (i) The embeddings carry significant information, with a rank information coefficient of about 0.07 and a t-statistic near 2.5. (ii) The nonlinear embedding does not improve on the linear one. (iii) The PCA portfolio with derivatives strategies returns 6.28% a year against 3.09% for the HSI TR, an active return of 3.05% with an information ratio of 0.46. Stock selection contributes 2.93%, and derivatives strategies contribute 0.98%. The forecast is significant, but its transfer into active return is not, and neither are the derivatives strategies, whose gains also depend on pricing options at the HSI Volatility Index (VHSI).

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 sets out the embeddings, return prediction, Black-Litterman portfolio construction, and derivatives strategies. Section 4 reports the results, and Section 5 concludes.

2 Data

The HSI has expanded rapidly over the last decades, from an index dominated by financials and property to a broader, technology-heavy benchmark. The investment universe on each date therefore consists of the stocks that are HSI constituents on that date. A stock enters the universe on the effective date of its inclusion and leaves on the effective date of its removal, so no date uses a later list.

Daily open, high, low and close prices, adjusted closes and volumes come from Yahoo Finance for the stocks that were HSI members at some date between January 2005 and August 2026. We treat a volume of zero as missing and carry the previous volume forward, since most such records fall on half-day sessions, and a day with zero volume and an unchanged close indicates a suspension. Where the data of the provider are plainly wrong, namely one spin-off adjustment and nonpositive adjusted closes, the prices of the stock on that day are set to missing. Seven codes are dropped, four without Yahoo data, among them Hang Seng Bank (0011.HK), a constituent until January 2026, and three that Yahoo now assigns to another company. Market data include the HSI, the VHSI, the Tracker Fund of Hong Kong (2800.HK) as a proxy for the HSI TR, and the USD/HKD rate. The risk-free rate is approximated by the one-month Hong Kong Interbank Offered Rate (HIBOR) fixing R_t^f published by the Hong Kong Monetary Authority, and cash accrues on the actual/365 basis at the daily return $r_t^f = \ln(1 + R_{t-1}^f \ell_t/365)$, with ℓ_t the number of calendar days since the previous trading day.

3 Methodology

Figure 1 summarizes the strategy's monthly cycle. At each rebalance, the past year of each stock is embedded in a low-dimensional representation. A pooled regression then maps the embedding to a forecast of the return over the next month. The forecasts enter portfolio construction via the Black-Litterman model. Finally, index derivatives are added to the resulting stock portfolio.

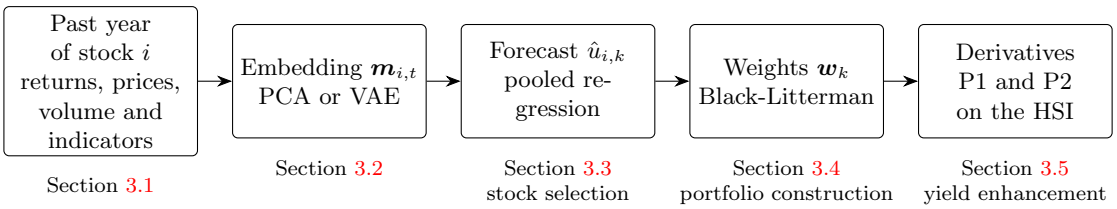


Figure 1. Monthly cycle of the strategy at rebalance τ_k.

Let $P_{i,t}$ be the adjusted close of stock i on trading day t and $r_{i,t} = \ln(P_{i,t}/P_{i,t-1})$ its return. The portfolio is rebalanced at the close of the first trading day of each month from September 2021 to August 2026, the dates $\tau_1, \ldots, \tau_{60}$, and every estimate used at τ_k relies on data up to the previous trading day $\tau_k - 1$.

3.1 Price path representation

The input of stock i on day t is the window $X_{i,t} \in \mathbb{R}^{L\times 16}$ of its $L = 240$ most recent trading days, the sequence length of Fischer and Krauss (2018). Each day includes 16 daily features adapted from Wu et al. (2026, Table 1), which are the return, open, high and low prices, volume and eleven technical indicators, transformed to be comparable across stocks, as listed in Table A.1. Both embeddings are fitted to describe the cumulative path of the standardized returns in the window,

$$c_{i,t,j} = \frac{1}{\sqrt{L}} \sum_{g=1}^{j} \frac{r_{i,t-L+g}}{s_{i,t}}, \qquad j = 1, \ldots, L, \tag{1}$$

where $s_{i,t}$ is the standard deviation of the returns in the window. The path $\boldsymbol{c}_{i,t} = (c_{i,t,1}, \ldots, c_{i,t,L})$ records when the gains and losses occurred, and its end value $c_{i,t,L}$ is the return over the window scaled by its volatility.

3.2 Dimension reduction by PCA and the VAE

This study examines two embeddings, PCA and the VAE. PCA is linear and takes the return path alone as input, whereas the VAE is nonlinear and takes the whole window, so it can find a low-dimensional curved surface on which the paths lie and draw on the technical indicators. Both are refitted annually on the first trading day of each September from 2021 to 2025, on the windows before the refit except those of the final 252 trading days, which the VAE holds out for validation.

The linear embedding $\boldsymbol{m}_{i,t} \in \mathbb{R}^2$ consists of the scores of $\boldsymbol{c}_{i,t}$ on its first two principal components. Two components carry 89.1% to 89.4% of the path variance at every refit, compared with 3.4% to 3.5% for the third, and they match the VAE dimension $d = 2$, so the embedding size is fixed for comparison.

The VAE (Kingma and Welling, 2014) encodes the window $X_{i,t}$ with an LSTM into the mean $\boldsymbol{\mu}_{i,t}$ and the log variance $\ln \boldsymbol{\sigma}^2_{i,t}$ of a Gaussian posterior over a latent $\boldsymbol{z} \in \mathbb{R}^d$, and a second LSTM

decodes a draw $\boldsymbol{z} = \boldsymbol{\mu}_{i,t} + \boldsymbol{\sigma}_{i,t} \odot \boldsymbol{\varepsilon}$, with $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, I)$, into an estimate $\hat{\boldsymbol{c}}_{i,t}$ of the path. The network minimizes the negative evidence lower bound for a Gaussian likelihood of variance one half,

$$\mathcal{L}_{i,t} = \sum_{j=1}^{L} (c_{i,t,j} - \hat{c}_{i,t,j})^2 + \mathrm{KL}\left(N(\boldsymbol{\mu}_{i,t}, \operatorname{diag} \boldsymbol{\sigma}_{i,t}^2) \,\|\, N(\mathbf{0}, I)\right), \tag{2}$$

with KL the Kullback-Leibler divergence, and its embedding is the posterior mean $\boldsymbol{m}_{i,t} = \boldsymbol{\mu}_{i,t}$, with $d = 2$ or $d = 3$. All stocks share a single network, as in the pooled design of Fischer and Krauss (2018). Appendix A gives the architecture and the training.

3.3 Return prediction

Stock selection rests on a forecast of each stock's return relative to the others. The target is the excess return over the next month relative to the cross section, and a pooled regression on the embedding gives the forecast,

$$y_{i,t} = \sum_{j=1}^{21} \left(r_{i,t+j} - r^f_{t+j}\right), \qquad u_{i,t} = y_{i,t} - \bar{y}_t, \tag{3}$$

$$u_{i,t} = a + \boldsymbol{b}^\top \boldsymbol{m}_{i,t} + \nu_{i,t}, \qquad \hat{u}_{i,k} = \hat{a} + \hat{\boldsymbol{b}}^\top \boldsymbol{m}_{i,\tau_k - 1}, \tag{4}$$

where $\bar{y}_t$ is the mean of $y_{i,t}$ over the HSI members on day t, following the cross-sectional target of Fischer and Krauss (2018), $\nu_{i,t}$ is the residual, and a and $\boldsymbol{b}$ are estimated at each refit by least squares pooled over the member days whose target ends before the refit. The forecast ranks the stocks, and its ranking power is measured by the rank information coefficient (IC) and by the t-statistic of its mean over the five years.

3.4 Portfolio construction

Portfolio construction turns the forecasts into weights with the Black-Litterman model. The universe $\mathcal{U}_k$ at τ_k is the set of p_k HSI constituents with a year of complete returns and an embedding. It ranges from 57 to 93 stocks over the backtest. The prior is defined as the equilibrium return implied by the HSI as the market portfolio, and each view states that a stock beats its prior by its forecast relative to the universe mean $\bar{u}_k$. The uncertainty of the views is proportional to the prior covariance $\zeta \Sigma_k$, with $\zeta > 0$ a scale that cancels from the posterior (He and Litterman, 1999),

$$\Pi_{i,k} = 21\, \delta \operatorname{Cov}\left(r_{i,t},\, r_{\mathrm{HSI},t}\right), \qquad \boldsymbol{Q}_k = \boldsymbol{\Pi}_k + \hat{\boldsymbol{u}}_k - \bar{u}_k \mathbf{1}, \qquad \Omega_k = \operatorname{diag}(\zeta \Sigma_k), \tag{5}$$

where $\delta = 2.5$ is the risk aversion of He and Litterman (1999), $r_{\mathrm{HSI},t}$ the daily return of the HSI, and Σ_k the covariance matrix over one month, estimated from the 252 daily returns before τ_k with the noise eigenvalues of the correlation matrix cleaned (Appendix B). With the identity matrix as the pick matrix, the posterior mean is

$$\boldsymbol{\mu}_k^{\mathrm{BL}} = \boldsymbol{\Pi}_k + \Sigma_k \left(\Sigma_k + \operatorname{diag} \Sigma_k\right)^{-1} \left(\hat{\boldsymbol{u}}_k - \bar{u}_k \mathbf{1}\right). \tag{6}$$

Each stock portfolio maximizes the mean-variance objective under a long-only constraint and a single-name cap of 3%,

$$\boldsymbol{w}_k = \arg\max_{\boldsymbol{w}} \; \boldsymbol{w}^\top \boldsymbol{\mu}_k - \frac{\delta}{2} \boldsymbol{w}^\top \Sigma_k \boldsymbol{w} \quad \text{subject to} \quad \mathbf{1}^\top \boldsymbol{w} = 1, \;\; 0 \le w_i \le 0.03. \tag{7}$$

All proposed portfolios differ only in the expected returns $\boldsymbol{\mu}_k$ (Table 1). M1 uses the prior alone, so M1 against the HSI TR measures portfolio construction. M2 adds the views, so M2 against M1 measures stock selection. Every portfolio is rebalanced monthly and pays 15 basis points one way on turnover, and a suspended stock keeps its weight.

Portfolio	Expected returns $\boldsymbol{\mu}_k$	Role
EW	none, $w_i = 1/p_k$	naive benchmark
M1	$\boldsymbol{\Pi}_k$	portfolio construction without stock selection
M2-PCA	$\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the PCA forecast	stock selection with the linear embedding
M2-VAE	$\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the VAE forecast, $d = 2$	stock selection with the nonlinear embedding
M3	$\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the VAE forecast, $d = 3$	stock selection with a larger nonlinear embedding

Table 1. Stock portfolios. All except EW solve Equation (7) with the stated expected returns.

3.5 Derivatives strategies

The derivatives are written on the HSI, the benchmark index. Each position holds $n = \left\lfloor \frac{h\,\beta_k\,E}{50\,S} \right\rfloor$ contracts, where E is the value of the stock portfolio, β_k its ex-ante beta to the HSI over the 252 days before τ_k, S the HSI close, 50 the contract multiplier in HKD and h the coverage.

We set the base case to $h = 0.5$, which covers 49% of the stock portfolio on average, and report $h = 1$ in Appendix E. Positions are opened in the next monthly contract to expire, at the first rebalance and at each expiry. Options are priced by the model of Black and Scholes (1973) with a continuous dividend yield (Merton, 1973) at the VHSI level v_t, and every trade pays transaction costs. Appendix C sets out the contract terms, the costs and the cash held for margin. Four scenarios are compared: no derivatives, P1, P2, and P1 with P2.

The yield strategy P1 targets the volatility risk premium. Implied variance tends to exceed the variance later realized (Carr and Wu, 2009), and the seller of a delta-hedged option earns on average what the buyer loses (Bakshi and Kapadia, 2003). A covered call collects this premium but also sells equity exposure and holds an uncompensated timing position (Israelov and Nielsen, 2015), which lowers the expected return of a portfolio that aims to beat the index. P1 therefore sells n out-of-the-money (OTM) calls at the first listed strike at or above $1.04S$ and hedges them with $H_t = \mathrm{round}\left(\frac{-\Delta_t}{e^{(R_t^f - q)\vartheta_t}}\right)$ HSI futures reset at every close, where Δ_t is the delta of the short calls in index units, ϑ_t the time to expiry and q the dividend yield. At a constant volatility, the hedged position gains

$$G \approx 25\,n \sum_t \Gamma_t S_t^2 \left(\frac{(v_0/100)^2}{A} - r_{\mathrm{HSI},t+1}^2 \right) \tag{8}$$

in HKD, with Γ_t the gamma of one call, v_0 the VHSI at which the calls are sold and A the number of trading days per year. P1 therefore gains when the index moves less than the VHSI implied.

The protection strategy P2 limits losses in falling markets at a low cost. Puts are costly protection, with strongly negative expected returns (Coval and Shumway, 2001). Protective puts reduce drawdowns less than a static cut in exposure (Israelov, 2019), while collars earn less equity premium than their index (Israelov and Klein, 2016). P2 therefore holds a put spread collar only when the VHSI signals stress, in the spirit of the adjusted collars of Szado and Schneeweis (2010), and finances the protection within the position.

The strategy triggers when the VHSI closes above 30 and exits when it closes below 22. Each position buys n puts at the first listed strike K_1 at or below $0.95S$, sells n puts at the first listed strike K_2 at or below $0.90S$, and sells n calls at the first listed strike K_3 at or above the strike within $[1.05S,\ 1.10S]$ that sets the net premium to zero. Figure 2 shows the resulting payoff. The collar absorbs losses between K_1 and K_2 and gives up gains above K_3, and selling the lower put sells the option with the lower expected return (Coval and Shumway, 2001). The size is capped at entry so that the absolute delta does not exceed 15% of the net asset value.

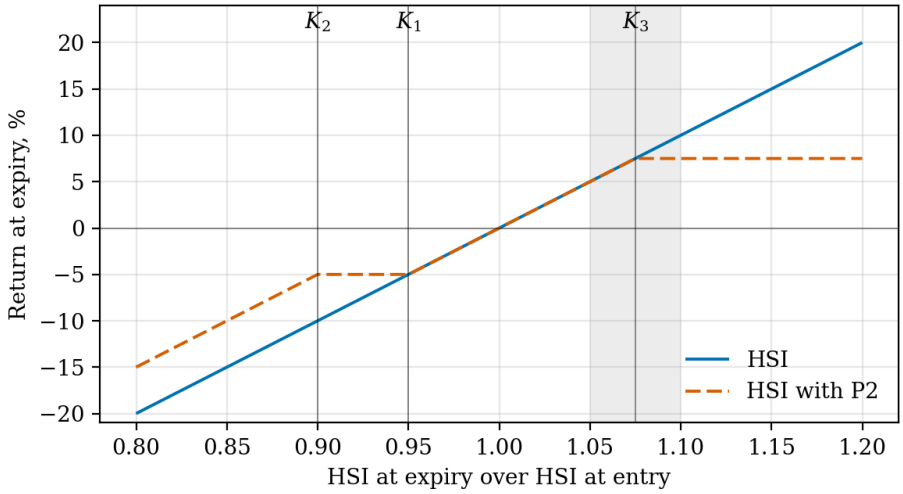


Figure 2. Return at expiry of the HSI and of the HSI with the P2 collar.

To isolate the design choices, we examine two variants. The first, P1 unhedged, holds the P1 calls without futures, which is a covered call. The second, P2 with the full trigger, holds P2 on a full trigger, which also turns on when the stock portfolio falls more than 5% below its running peak and turns off only when the VHSI is below 22 and the drawdown is within 2%.

4 Results

Section 4.1 examines and compares the forecasting performance of PCA and the VAE. Section 4.2 examines whether the forecast transfers to active return. Section 4.3 examines the profitability of derivatives strategies. Section 4.4 decomposes the final portfolio's outperformance by source. Performance is measured by annualized return, volatility, Sharpe ratio and maximum drawdown (MDD) and, against a benchmark, by active return, tracking error (TE) and information ratio (IR), as defined in Appendix D.

4.1 Predictive information in the embeddings

	Mean IC by year					All months		
Embedding	2021/22	2022/23	2023/24	2024/25	2025/26	Mean	t	IC > 0, %
PCA	0.094	0.054	0.136	0.035	0.040	0.072	2.47	63
VAE ($d = 2$)	0.098	0.069	0.138	0.039	0.047	0.078	2.66	63
VAE ($d = 3$)	0.084	0.050	0.026	0.024	0.040	0.045	1.49	58

Table 2. Rank IC of each forecast, averaged within each year from September to August and over all 60 months, with the t-statistic of the mean and the share of months with a positive IC.

Both embeddings carry predictive information about next-month returns (Table 2). The forecasts from PCA and from the VAE with $d = 2$ have mean ICs of 0.072 and 0.078, with t-statistics of 2.47 and 2.66. The VAE with $d = 3$ is weaker and not significant. The pooled regression explains

at most 0.13% of the variance of monthly relative returns, so the information lies in the ranking of stocks rather than in return levels.

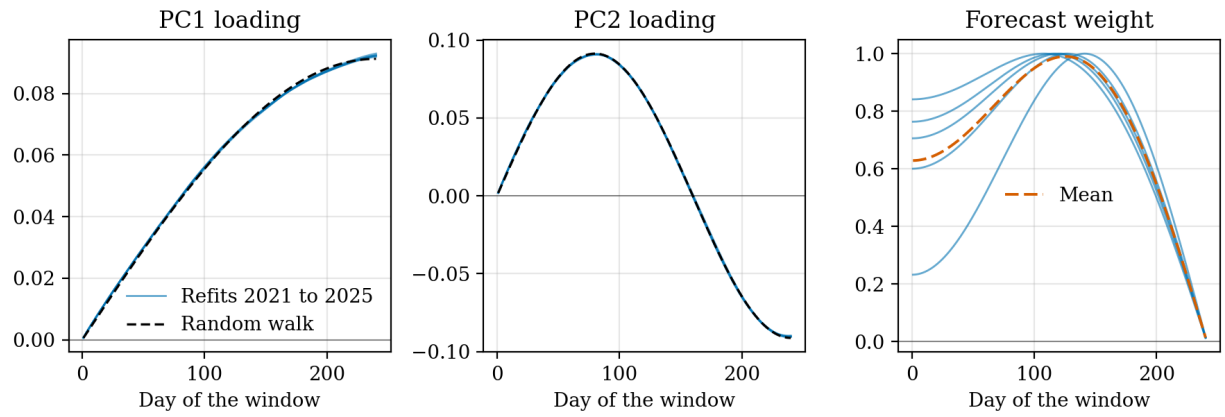


Figure 3. Loadings of the first two principal components at each refit from 2021 to 2025 with the eigenvectors of a random walk, and the forecast weight η_g relative to the largest weight.

Figure 3 shows what the linear embedding measures, and we find that the standardized return path of an HSI stock behaves like a random walk. At every refit, the loadings of the first two principal components coincide with the eigenvectors of a random walk, and their shares of the variance, about 80% and 9%, match the 81% and 9% of a random walk. The first component weights the whole window and tracks the volatility-scaled return over the year, whereas the second contrasts the first two-thirds of the window with the last third and so measures whether the gains came early or late.

Because the path is cumulative, the forecast is a weighted sum of the standardized daily returns,

$$\hat{u}_{i,k} = \text{const} + \sum_{g=1}^{L} \eta_g \frac{r_{i,\tau_k-1-L+g}}{s_{i,\tau_k-1}}, \qquad \eta_g = \frac{1}{\sqrt{L}} \sum_{j=g}^{L} \left(\hat{b}_1 \chi_{1,j} + \hat{b}_2 \chi_{2,j} \right), \tag{9}$$

where $\chi_{1,j}$ and $\chi_{2,j}$ are the loadings of the two components and $\hat{b}_1$ and $\hat{b}_2$ their coefficients. As shown in the right panel of Figure 3, the forecast weights are largest for returns from three to nine months before the forecast and small for the most recent month, so the forecast reflects momentum over the past year (Jegadeesh and Titman, 1993).

Random-walk behavior also explains why the nonlinear embedding does not outperform the linear one. Since a random walk is close to a Gaussian process, whose covariance describes it fully, a nonlinear embedding has little to find beyond the directions of largest variance, and a VAE with linear maps recovers exactly the principal subspace (Lucas et al., 2019). The technical indicators do not alter this, because the VAE is trained to rebuild the return path and most indicators are functions of the same closing prices. Hence, the PCA scores explain the VAE embedding with an R^2 above 0.9 in four of the five refits (Table A.2). The monthly ICs of the two forecasts have a correlation of 0.99, although the IC of the VAE is marginally higher in every year. With $d = 3$, one dimension is inactive in four refits, which adds no information and lowers the IC.

4.2 Stock portfolios

The backtest shows that stock selection turns the forecast into outperformance, whereas portfolio construction alone does not. EW and M1 underperform the HSI TR, while M2-PCA and M2-VAE

Portfolio	Return	Volatility	Sharpe ratio	Max drawdown	Turnover	Avg. # holdings
EW	2.0	24.1	−0.04	−39.8	8.0	78
M1	2.2	25.3	−0.03	−43.0	18.6	42
M2-PCA	5.3	23.9	0.09	−40.6	20.4	40
M2-VAE	4.7	24.0	0.07	−40.6	21.8	40
M3	3.8	24.0	0.03	−40.7	25.6	40
HSI TR	3.1	24.6	0.00	−41.7		

Table 3. Performance of the stock portfolios from September 2021 to August 2026.

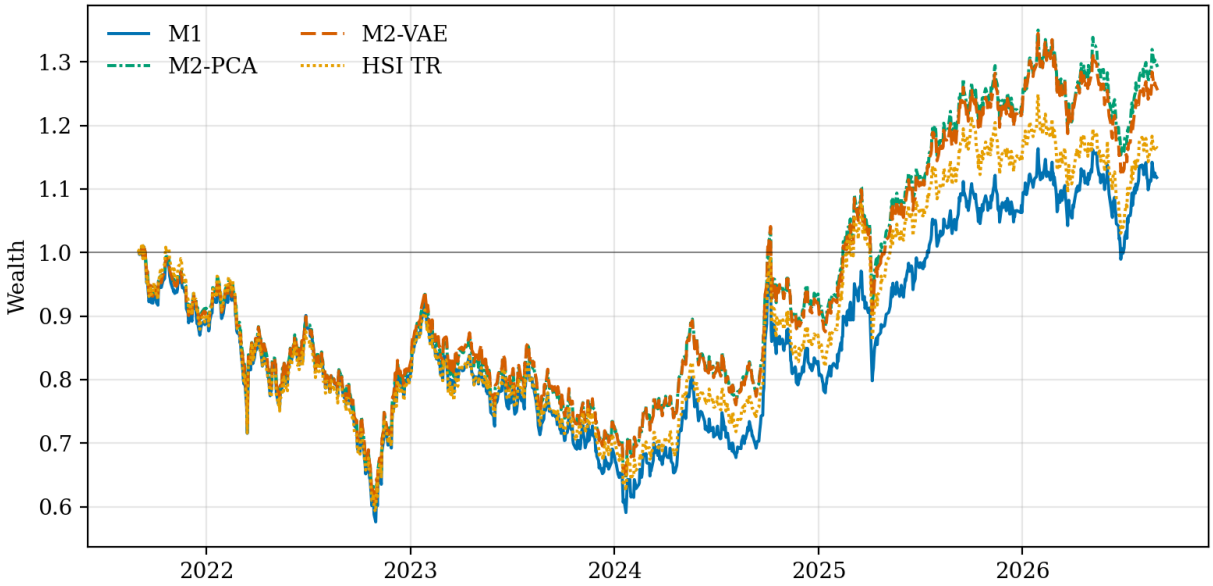


Figure 4. Wealth of M1, M2-PCA, M2-VAE and the HSI TR.

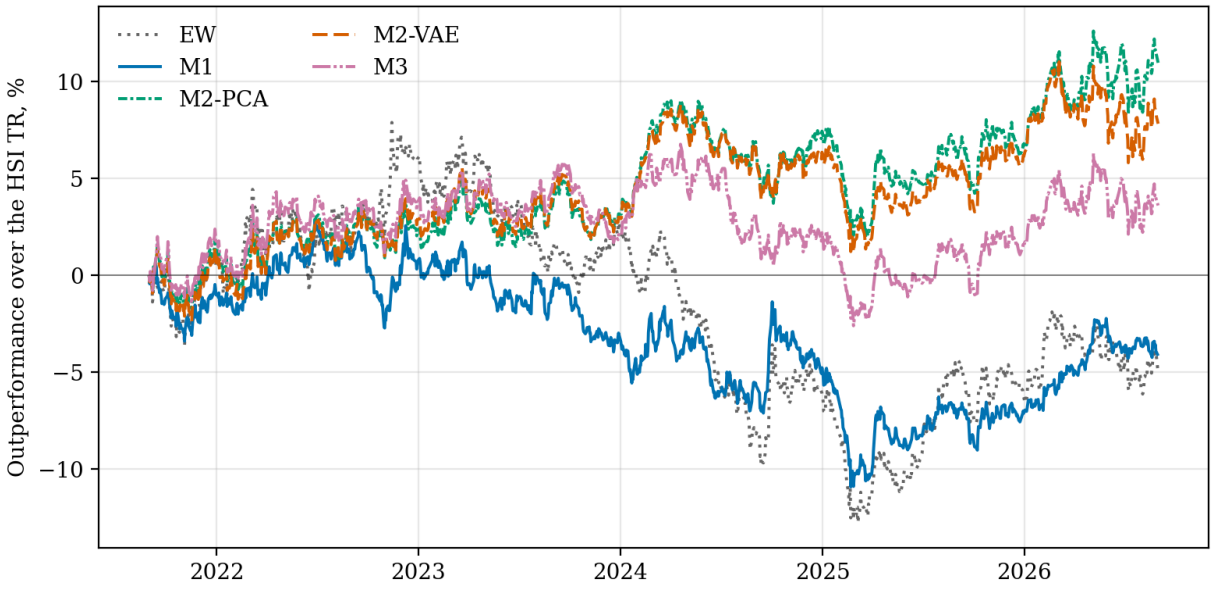


Figure 5. Outperformance of each stock portfolio over the HSI TR, in %.

outperform it with lower volatility (Table 3).

The outperformance of the M2 portfolios arises from stock selection. M2-PCA beats M1 in every year, and M2-VAE also beats it in four of the five years (Table 5). Against M1, which isolates stock selection, M2-PCA and M2-VAE earn active returns of 2.9% and 2.4% with t-statistics of 1.70 and 1.40, respectively (Table 4). The outperformance of M2-PCA and M2-VAE increases steadily over time rather than in any specific market environment (Figures 4 and 5). Furthermore, the M2 portfolios maintain a beta of 0.95 to the HSI, so the gain is not a market bet.

The difference between M2-PCA and M2-VAE, 0.57% a year with a t-statistic of 1.24, is not significant. The two portfolios share 95% of their weights, as expected from forecasts that rank stocks alike. M3 inherits the weaker forecast of its VAE and earns less than the M2 portfolios.

	Against the HSI TR					Against M1		
Portfolio	Beta	Active	TE	IR	t	Active	IR	t
EW	0.94	−1.0	6.7	−0.16	−0.35			
M1	1.01	−0.8	5.2	−0.16	−0.36			
M2-PCA	0.95	+2.1	5.2	+0.40	+0.89	+2.9	+0.76	+1.70
M2-VAE	0.95	+1.5	5.2	+0.29	+0.65	+2.4	+0.62	+1.40
M3	0.95	+0.7	5.2	+0.14	+0.30	+1.6	+0.39	+0.88

Table 4. Active performance of the stock portfolios against the HSI TR and against M1. Active return and TE are in % a year.

Portfolio	2021/22	2022/23	2023/24	2024/25	2025/26
EW	+1.3	−0.6	−9.1	+5.2	−0.6
M1	+0.9	−2.1	−4.6	−2.1	+3.1
M2-PCA	+1.6	+1.3	+3.1	0.0	+4.3
M2-VAE	+1.4	+1.9	+2.5	−1.3	+2.4
M3	+2.3	+1.5	−1.5	−2.4	+2.4
HSI TR	−21.0	−3.8	2.3	43.2	4.7

Table 5. Outperformance of the stock portfolios over the HSI TR in each September-to-August year, in %, with the HSI TR return in the last row.

4.3 Derivatives strategies

Table 6 reports the derivatives strategies on M2-PCA and M2-VAE. Since their effect is the same on all portfolios to within 0.07% a year, the discussion below refers to M2-PCA.

Yield enhancement comes from hedging. Without futures, the covered call (P1 unhedged) lowers volatility by selling index exposure but also reduces returns, whereas P1 raises returns by 0.73% a year at unchanged volatility, with a t-statistic of 1.43. This finding is consistent with a volatility risk premium in HSI options, as the VHSI exceeded the subsequently realized volatility in 44 of the 60 months.

Conditioning the collar on the VHSI keeps protection costs low. P2 is held on 31% of days, raises the return by 0.31% a year and lowers the MDD from 39.2% to 35.5%, whereas the full trigger (P2, full trigger) holds the collar 85% of the time and lowers the return by 2.00% a year. Figure 6 shows that the gain of P2 arises in the decline of 2022. The collar was held from March

		Performance				Over none		Over HSI TR		
Portfolio	Scenario	Return	Vol.	Sharpe	MDD	Gain	t	Active	IR	Days
M2-PCA	None	5.24	22.8	0.10	−39.2			+2.06	0.38	
	P1	5.97	22.9	0.13	−38.6	+0.73	1.43	+2.75	0.51	
	P2	5.55	21.1	0.12	−35.5	+0.31	0.21	+2.36	0.36	31
	P1 and P2	6.28	21.2	0.15	−34.9	+1.04	0.68	+3.05	0.46	31
	P1 unhedged	5.02	19.9	0.10	−34.3	−0.22	−0.10	+1.85	0.23	
	P2, full trigger	3.24	19.7	0.01	−35.8	−2.00	−1.04	+0.15	0.02	85
M2-VAE	None	4.66	22.9	0.07	−39.2			+1.52	0.28	
	P1	5.39	23.0	0.10	−38.6	+0.72	1.42	+2.20	0.41	
	P2	4.97	21.2	0.09	−35.5	+0.31	0.21	+1.81	0.28	31
	P1 and P2	5.70	21.3	0.12	−34.9	+1.04	0.68	+2.50	0.38	31
	P1 unhedged	4.45	20.0	0.07	−34.2	−0.22	−0.10	+1.31	0.17	
	P2, full trigger	2.64	19.8	−0.02	−35.8	−2.02	−1.05	−0.43	−0.06	86

Table 6. Derivatives strategies on M2-PCA and M2-VAE at coverage $h = 0.5$. Gain is the difference in annual return from no derivatives, active return and IR are measured against the HSI TR, and days is the share of days on which P2 is held, in %. P1 unhedged holds the calls of P1 without futures, and P2 on the full trigger adds a drawdown condition.

2022 to April 2023, and its short call gave up the whole gain in the rebound from November 2022 before the collar closed ahead in April 2023. Its three later spells, in 2023 to 2025, left the relative value almost unchanged.

Together, P1 and P2 improve the annualized return by 1.04% and reduce both the volatility and the MDD.

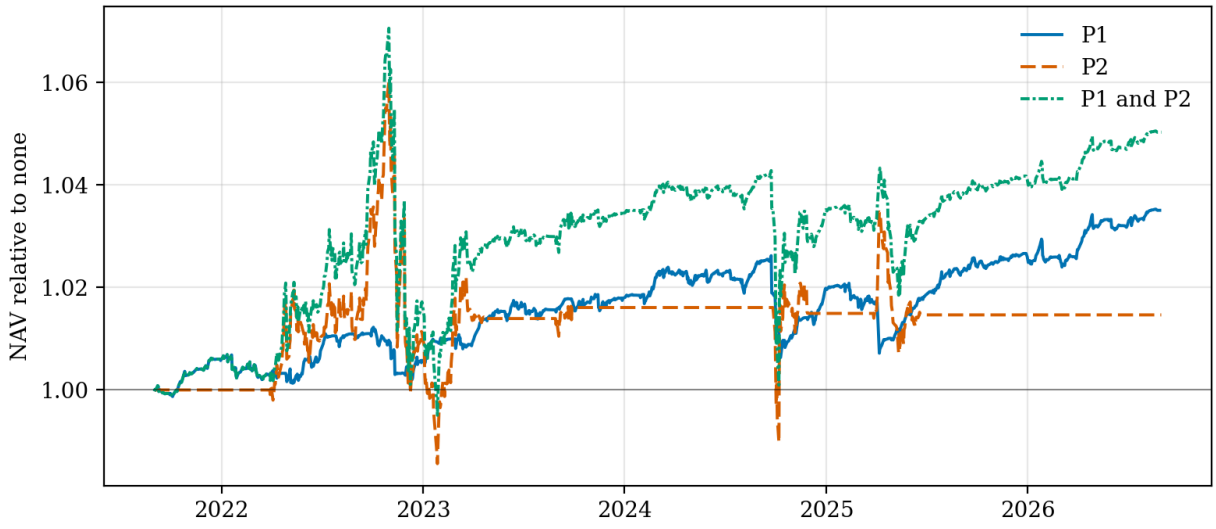


Figure 6. Net asset value of M2-PCA with derivatives strategies relative to that without derivatives.

4.4 Sources of outperformance

Table 7 decomposes the active return of the final portfolio, M2-PCA with P1 and P2, which returns 6.28% a year against 3.09% for the HSI TR with a lower volatility and a smaller drawdown. Each

Source	Comparison	M2-PCA			M2-VAE		
		Active	IR	t	Active	IR	t
Portfolio construction	M1 against the HSI TR	−0.85	−0.16	−0.36	−0.85	−0.16	−0.36
Stock selection	M2 against M1	+2.93	+0.76	+1.70	+2.35	+0.62	+1.40
Derivatives, P1	P1 against none	+0.69	+0.64	+1.43	+0.69	+0.64	+1.42
Derivatives, P2	P1 and P2 against P1	+0.29	+0.09	+0.21	+0.30	+0.10	+0.21
Total	P1 and P2 against the HSI TR	+3.05	+0.46	+1.03	+2.50	+0.38	+0.86

Table 7. Sources of the active return of M2-PCA and M2-VAE with P1 and P2 against the HSI TR at $h = 0.5$. Each step is measured against the previous one, and the steps add up to the total except for the cash held for margin, which changes the active return by −0.02% for M2-PCA and +0.01% for M2-VAE. None is the portfolio without derivatives.

row measures one step against the previous one: portfolio construction alone loses 0.85% a year against the HSI TR, stock selection adds 2.93%, and derivatives add 0.98%. Stock selection from the embedding forecast is the main source of the outperformance. Although the forecast itself is significant, its transfer into active return is not, and neither are the derivatives strategies. The same pattern holds for M2-VAE.

5 Conclusion

In this study, we examine (i) whether embeddings of the past year of returns and technical indicators of HSI stocks carry information about their returns over the next month, (ii) whether a nonlinear embedding improves on a linear one, and (iii) what the embeddings deliver in a long-only portfolio with index derivatives. We embed each stock's trading history with PCA and an LSTM VAE, forecast returns with a pooled regression on the embedding, construct weights with the Black-Litterman model, and enhance yield with delta-hedged short calls and a put spread collar held when the VHSI is high.

The embeddings carry significant information about returns, and the forecast they produce is momentum over the past year. The VAE and PCA perform similarly, because the standardized return path behaves like a random walk that leaves little structure beyond its principal components. From September 2021 to August 2026, the PCA portfolio with derivatives strategies outperforms the HSI TR by 3.05% a year with lower volatility and drawdown. The outperformance mainly stems from stock selection, while the hedged calls enhance yield and the conditional collar protects at little cost. However, neither the transfer of the forecast into active return nor the gain from the derivatives is statistically significant.

Pricing at the VHSI is a limitation of this study. The VHSI is computed from HSI options across strikes using the VIX method (Hang Seng Indexes Company Limited, 2023), so its square approximates a 30-day variance swap rate (Carr and Wu, 2009) rather than the implied volatility of any traded strike. Under a negative skew, OTM calls trade below the VHSI and OTM puts above it. The backtest gain from the derivatives disappears if calls are priced about 1.6 volatility points below the VHSI and puts about 1.6 points above it.

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A The input features and the VAE

Table A.1 lists the input features, Figure A.1 draws the architecture of the VAE, and Table A.2 reports the diagnostics of its embeddings at each refit.

The encoder and decoder have two LSTM layers with 32 hidden units and dropout 0.25. Training uses Adam with a learning rate of 0.001, batches of 256 windows and gradient norm clipping at one, and stops after ten epochs without improvement in the loss on the windows of the final 252 trading days before the refit, which are held out for validation, with a ceiling of 300 epochs.

Feature	Definition
Open	$P^{\mathrm{o}}_t / P^{\mathrm{c}}_{t-1} - 1$
High	$P^{\mathrm{h}}_t / P^{\mathrm{c}}_{t-1} - 1$
Low	$P^{\mathrm{l}}_t / P^{\mathrm{c}}_{t-1} - 1$
Return	$r_t = \ln(P_t / P_{t-1})$
Volume	$\ln \mathcal{V}_t$
True range	$\mathrm{TR}_t / P^{\mathrm{c}}_t$, with $\mathrm{TR}_t = \max\left(P^{\mathrm{h}}_t - P^{\mathrm{l}}_t, \lvert P^{\mathrm{h}}_t - P^{\mathrm{c}}_{t-1}\rvert, \lvert P^{\mathrm{l}}_t - P^{\mathrm{c}}_{t-1}\rvert\right)$
Average true range	mean of TR_t over 20 days, over P^{c}_t
Simple moving averages	P^{c}_t over its mean over 5, 10 or 20 days, less one
Exponential moving averages	P^{c}_t over its exponential average with span 5, 10 or 20 days, less one
Stochastic %K	$(P^{\mathrm{c}}_t - \min P^{\mathrm{l}}) / (\max P^{\mathrm{h}} - \min P^{\mathrm{l}})$ over 14 days
Stochastic %D	mean of %K over 3 days
Relative strength index	average rise of P^{c} over its average rise plus average fall, smoothing 1/14

Table A.1. The sixteen daily input features of the VAE after Wu et al. (2026, Table 1) on day t. P^{o}, P^{h}, P^{l} and P^{c} are the split-adjusted open, high, low and close, P is the dividend- and split-adjusted close, $\mathcal{V}$ is the volume. Within each window of L days, the return is divided by its standard deviation and the log volume is demeaned, and the other features are standardized with means and deviations from the training days.

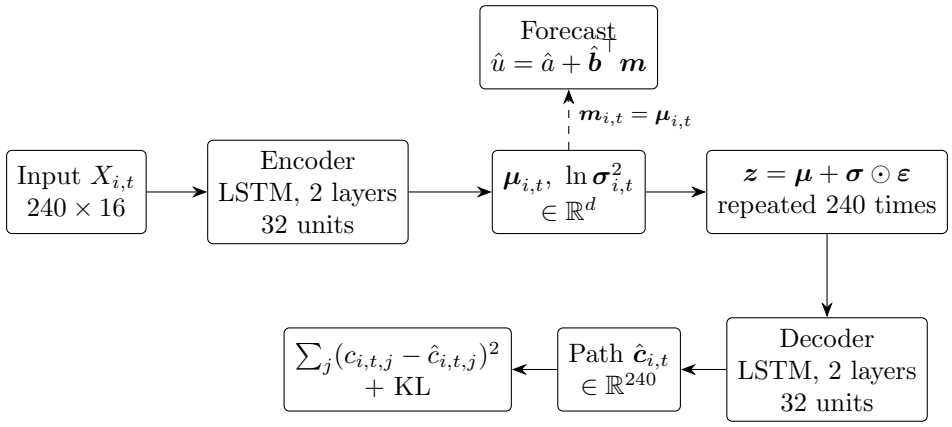


Figure A.1. Architecture of the VAE. The encoder reads 240 days of sixteen features and returns the mean and log variance of the latent, and the decoder receives a draw of the latent at each of 240 steps and rebuilds the cumulative path of standardized returns. The posterior mean $\boldsymbol{m}_{i,t}$ is the embedding that enters the forecast.

		Active units		R^2 on PCA scores	
Refit	Training windows	$d = 2$	$d = 3$	Dimension 1	Dimension 2
2021	154,401	2	2	0.91	0.97
2022	167,233	2	2	0.91	0.97
2023	182,816	2	3	0.94	0.97
2024	201,166	2	2	0.68	0.91
2025	220,792	2	2	0.92	0.96

Table A.2. Diagnostics of the VAE embeddings at each refit. Active units count the latent dimensions whose posterior mean has a variance across windows above 0.01 (Burda et al., 2016). R^2 is that of each dimension of the embedding of the VAE with $d = 2$ regressed on the two PCA scores of the same windows.

B Covariance estimation

The covariance matrix at τ_k is estimated from the 252 daily returns before τ_k of the stocks in $\mathcal{U}_k$. Let D_k be the diagonal matrix of their variances and $C_k = \Xi_k \Lambda_k \Xi_k^\top$ their correlation matrix, with eigenvectors Ξ_k and eigenvalues $\Lambda_k = \mathrm{diag}(\lambda_1, \ldots, \lambda_{p_k})$ in decreasing order. The eigenvalues above the upper edge of the Marchenko and Pastur law (Marchenko and Pastur, 1967) are kept as factors, at least one and at most four, and the others are replaced by their mean $\bar{\lambda}_k$ (Laloux et al., 2000),

$$f_k = \min\left(\max\left(\#\left\{\lambda \in \{\lambda_1, \ldots, \lambda_{p_k}\} : \lambda > \left(1 + \sqrt{p_k/252}\right)^2\right\}, 1\right), 4\right), \tag{10}$$

$$\tilde{\Lambda}_k = \mathrm{diag}\left(\lambda_1, \ldots, \lambda_{f_k}, \bar{\lambda}_k, \ldots, \bar{\lambda}_k\right). \tag{11}$$

The covariance matrix over one month is

$$\Sigma_k = 21\, D_k^{1/2}\, \tilde{C}_k\, D_k^{1/2}, \tag{12}$$

where $\tilde{C}_k$ is $\Xi_k \tilde{\Lambda}_k \Xi_k^\top$ rescaled to a unit diagonal.

C Derivatives contracts and costs

The portfolio starts with USD 100 million converted to HKD and holds the stock portfolio E_t, cash B_t earning r_t^f and derivatives worth O_t, so that its net asset value is $V_t = E_t + B_t + O_t$. At each rebalance the cash share is set in proportion to the initial margin of the futures,

$$\phi_k = \rho\, h\, \beta_k, \qquad \rho = \frac{117{,}705}{50\, S_{\mathrm{M}}}, \tag{13}$$

where ρ is the initial margin of HK\$117,705 for one HSI futures contract as a share of its notional value at the index level S_{M} of 2 March 2026. The stock portfolio is then reset to $E = (1 - \phi_k)V$ at a cost of 15 basis points. Every scenario holds the same cash share, so the scenarios differ only by their derivatives.

HSI options and futures have a multiplier of HK\$50 and expire on the second-to-last trading day of the month, and listed strikes are 100 points apart below 20,000 and 200 points apart above. Options are valued at the VHSI level v_t, the HIBOR fixing and a dividend yield q of 3%, the average trailing yield of 2800.HK from 2014 to August 2021 rounded, and are cash settled at the closing

level on expiry at no cost. Each option trade pays half of the widest spread the exchange permits its market makers to quote,

$$\psi(x) = \begin{cases} \max(30,\, 0.1x) & x \le 750, \\ 75 & x > 750, \end{cases} \tag{14}$$

in index points for a premium of x points, and each option or futures contract pays an exchange fee and levy of HK\$10.54. The futures price $F_t = S_t e^{(R_t^f - q)\vartheta_t}$ settles to cash every day, and each futures trade also pays a half spread of one index point, the minimum tick.

D Performance measures

Let r_t^{p} and r_t^{b} be the daily returns of a portfolio and of a benchmark, $t = 1, \ldots, T$, over a sample of Y years with A trading days per calendar year. For a daily series x_t, write

$$s(x) = \{(T-1)^{-1} \sum_t (x_t - \bar{x})^2\}^{1/2}, \qquad \bar{x} = T^{-1} \sum_t x_t \tag{15}$$

With the excess returns $e_t^{\mathrm{p}} = r_t^{\mathrm{p}} - r_t^f$ and $e_t^{\mathrm{b}} = r_t^{\mathrm{b}} - r_t^f$,

$$\tilde{r}_t = r_t^{\mathrm{p}} - r_t^{\mathrm{b}}, \qquad W_t = \exp(\sum_{s \le t} r_s^{\mathrm{p}}) \tag{16}$$

$$\begin{aligned} \text{Return} &= e^{A\bar{r}^{\mathrm{p}}} - 1, & \text{Vol} &= \sqrt{A}\, s(r^{\mathrm{p}}), \\ \text{SR} &= \sqrt{A}\, \frac{\bar{e}^{\mathrm{p}}}{s(e^{\mathrm{p}})}, & \text{MDD} &= \min_t \left(\frac{W_t}{\max\left(1, \max_{s \le t} W_s\right)} - 1 \right), \\ \text{Active} &= A\, \bar{\tilde{r}}, & \text{TE} &= \sqrt{A}\, s(\tilde{r}), \\ \text{IR} &= \frac{\text{Active}}{\text{TE}}, & t_{\text{IR}} &= \text{IR}\sqrt{Y}, \\ \beta^{\mathrm{p}} &= \frac{\sum_t \left(e_t^{\mathrm{p}} - \bar{e}^{\mathrm{p}}\right)\left(e_t^{\mathrm{b}} - \bar{e}^{\mathrm{b}}\right)}{\sum_t \left(e_t^{\mathrm{b}} - \bar{e}^{\mathrm{b}}\right)^2}, \end{aligned}$$

where Return, Vol and SR are the annualized return, volatility and Sharpe ratio, Active is the active return, t_{IR} its t-statistic and β^{p} the beta. A two-sided test at 5% requires $|t_{\text{IR}}| \ge 1.96$, that is $|\text{IR}| \ge 1.96/\sqrt{Y} = 0.88$ for the $Y = 4.99$ years of the backtest.

E Derivatives strategies at full coverage

Table E.1 doubles the coverage to $h = 1$. On M2-PCA, P1 adds 1.38% a year, P2 adds 0.34% and the two strategies together add 1.72%, for an active return of 3.66% against the HSI TR with an IR of 0.50. The contribution of P1 scales with its size, while the delta cap binds at all 19 entries of P2, which therefore changes little.

		Performance				Over none		Over HSI TR		
Portfolio	Scenario	Return	Vol.	Sharpe	MDD	Gain	t	Active	IR	Days
M2-PCA	None	5.21	21.8	0.10	−37.7			+2.04	0.36	
	P1	6.60	22.0	0.16	−36.6	+1.38	1.42	+3.35	0.57	
	P2	5.55	20.0	0.12	−33.9	+0.34	0.23	+2.36	0.33	31
	P1 and P2	6.94	20.2	0.19	−32.7	+1.72	0.99	+3.66	0.50	31
M2-VAE	None	4.66	21.9	0.07	−37.7			+1.51	0.27	
	P1	6.05	22.1	0.13	−36.5	+1.39	1.42	+2.83	0.49	
	P2	5.00	20.1	0.10	−33.9	+0.34	0.23	+1.84	0.26	31
	P1 and P2	6.39	20.3	0.16	−32.6	+1.73	0.99	+3.15	0.44	31

Table E.1. Derivatives strategies on M2-PCA and M2-VAE at coverage $h = 1$, with columns as in Table 6.